

COLOUR BASED PRODUCT SORTING

SMART RETAIL SHELF

ABSTRACT:

The Smart Retail Shelf is an advanced automated inventory management system designed to improve efficiency in retail environments. In traditional retail stores, monitoring product availability is done manually, which often leads to human errors, delayed restocking, and poor customer satisfaction. This project introduces a smart solution using Internet of Things (IoT) technology to continuously monitor the stock level on retail shelves.

The system uses load cell sensors to measure the weight of products placed on the shelf. This weight data is processed by a microcontroller such as Arduino or ESP32. When the stock level falls below a predefined threshold, the system triggers an alert through a buzzer and sends real-time notifications to a cloud platform or mobile application. This enables store managers to take immediate action.

1. INTRODUCTION:

Retail industries are rapidly evolving with the adoption of automation and smart technologies. One of the major challenges faced by retailers is maintaining accurate inventory levels on shelves. Customers often face inconvenience when products are unavailable, leading to dissatisfaction and loss of business.

Traditional inventory management systems rely heavily on manual inspection, barcode scanning, or periodic stock checking. These methods are time-consuming, inefficient, and prone to errors. To overcome these limitations, smart systems are required that can provide real-time monitoring and automation.

The Smart Retail Shelf is an IoT-based solution that integrates sensors, microcontrollers, and cloud computing to create an intelligent monitoring system. By continuously tracking the weight of items on the shelf, it can estimate the number of products available. The system automatically detects low stock conditions and alerts the store personnel.

This project demonstrates how modern embedded systems and IoT technologies can be applied to solve real-world retail problems, improving operational efficiency and customer satisfaction.

2. PROBLEM STATEMENT:

In current retail environments, several problems arise due to inefficient inventory management:

- Manual stock checking requires significant time and labor
- Human errors lead to incorrect inventory tracking

- Delayed restocking causes empty shelves
- Overstocking increases storage and maintenance costs
- Lack of real-time data prevents effective decision-making
- Customers leave stores due to unavailability of products

Problem Definition:

There is a need for an automated, reliable, and real-time inventory monitoring system that can detect stock levels and notify store managers instantly without human intervention.

3. PROPOSED SYSTEM:

The proposed Smart Retail Shelf system is designed to overcome the limitations of traditional methods by implementing an automated and intelligent solution.

System Description:

- A load cell sensor is placed under the shelf to measure weight
- The HX711 module amplifies sensor signals
- A microcontroller processes the data
- LCD displays real-time weight
- WiFi module sends data to cloud/server
- Alerts are generated when stock is low

Key Features:

- Continuous monitoring of product quantity
- Real-time alerts and notifications
- Remote access using IoT platform
- Low power consumption
- Scalable for multiple shelves

Advantages:

- Reduces manual effort
- Improves inventory accuracy
- Prevents stockouts
- Enhances customer satisfaction
- Enables smart decision-making

4. COMPONENTS REQUIRED:

Hardware Components (Detailed):

1. Microcontroller (Arduino / ESP32):
Acts as the brain of the system, processes sensor data and controls output devices.
2. Load Cell (Weight Sensor):
Measures the weight of products placed on the shelf.
3. HX711 Amplifier Module:
Converts small analog signals from load cell into digital values.
4. LCD Display (16x2):
Displays current weight and status messages.
5. Buzzer:
Provides audible alert when stock is low.
6. WiFi Module (ESP8266):
Enables internet connectivity for IoT communication.
7. Power Supply:
Provides required voltage to all components.
8. Connecting Wires & Breadboard:
Used for circuit connections.

Software Components (Detailed):

- Arduino IDE: Used for writing and uploading code
- Embedded C: Programming language
- ThingSpeak / Blynk: Cloud platform for data monitoring
- Serial Monitor: For debugging and output display

5. PROGRAM (EXPLANATION INCLUDED):

```
#include "HX711.h"
```

```
#define DOUT 3
```

```
#define CLK 2
```

```
HX711 scale;
```

```
int buzzer = 8;
```

```
float weight;
```

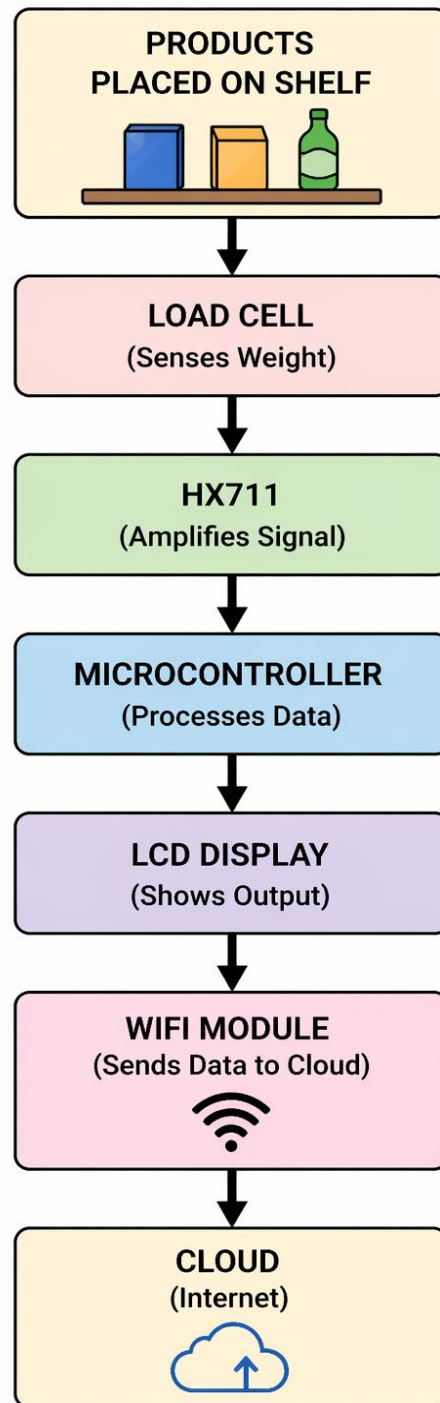
```
void setup() {
```

```
Serial.begin(9600);  
scale.begin(DOUT, CLK);  
pinMode(buzzer, OUTPUT);  
}  
void loop() {  
    weight = scale.get_units();  
  
    Serial.print("Weight: ");  
    Serial.println(weight);  
  
    if(weight < 50) {  
        digitalWrite(buzzer, HIGH);  
        Serial.println("Stock Low!");  
    } else {  
        digitalWrite(buzzer, LOW);  
    }  
  
    delay(1000);  
}
```

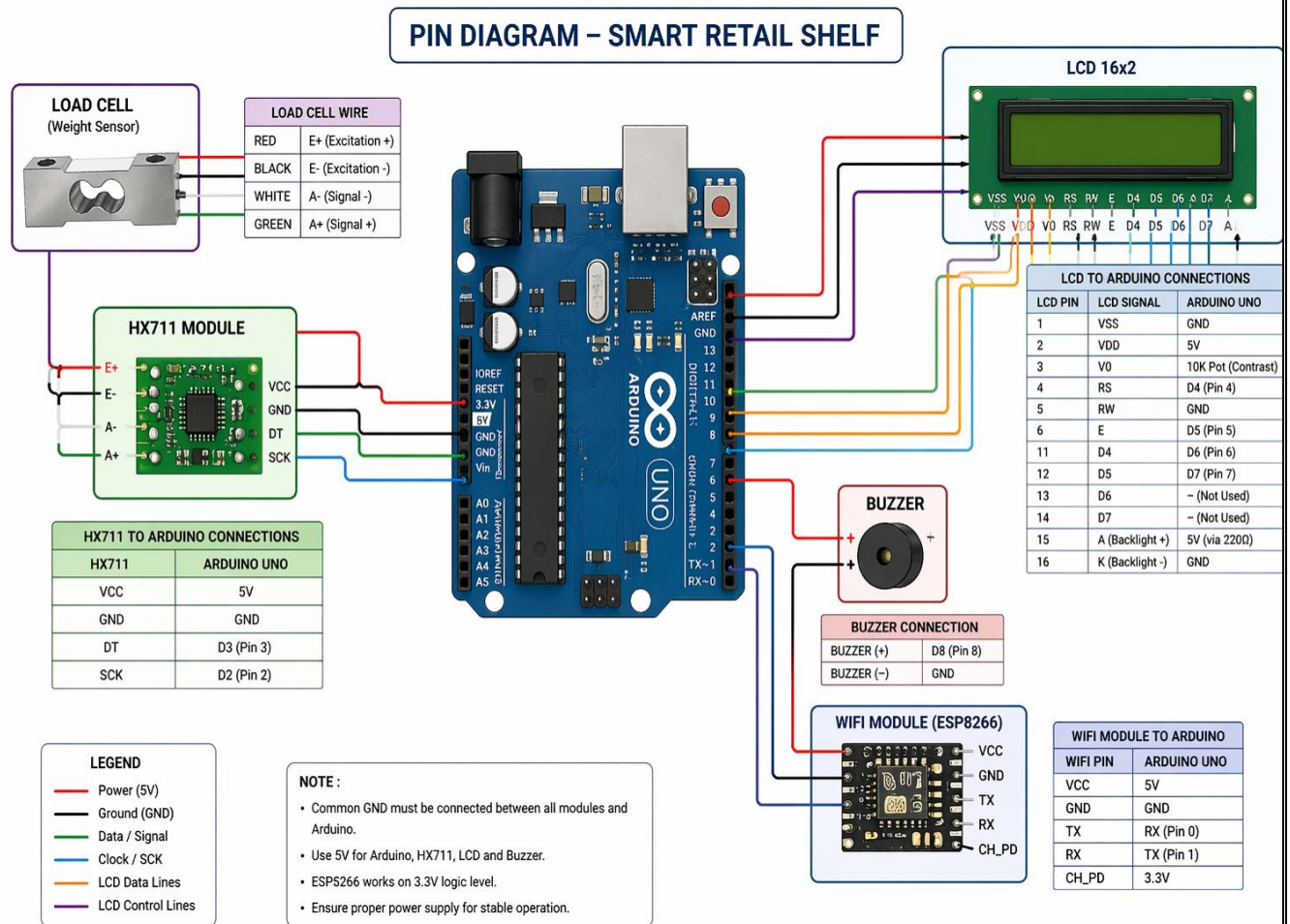
Program Explanation:

- HX711 library reads weight from load cell
- Weight is continuously monitored
- If weight < threshold → buzzer ON
- Data printed on serial monitor

6. BLOCK DIAGRAM :



7. PIN DIAGRAM :



8. OUTPUT:

Case 1: When weight is MORE than 50

Weight: 120.45

Weight: 118.30

Weight: 110.75

👉 Buzzer = OFF

👉 Status = Normal

Case 2: When weight is LESS than 50

Weight: 45.20

Stock Low!

Weight: 30.10

Stock Low!

Weight: 10.50

Stock Low!

👉 Buzzer = ON 🔊

👉 Alert message printed

9. WORKING PROCEDURE :

Step 1: Power ON the System

When the system is powered ON, all components such as the microcontroller, load cell, HX711 module, LCD display, buzzer, and WiFi module get initialized. The microcontroller starts executing the program, and initial messages may be displayed on the LCD indicating system readiness.

Step 2: Placement of Products on Shelf

Products are placed on the retail shelf. The shelf is mounted over the load cell sensor, so the total weight of the items is directly applied to the sensor.

Step 3: Load Cell Detects Weight

The load cell acts as a weight sensor. When products are placed on it, it produces a small analog electrical signal proportional to the applied weight. This signal is very weak and cannot be directly processed by the microcontroller.

Step 4: HX711 Amplifies and Converts Signal

The HX711 module is used to amplify the weak analog signal from the load cell. It also converts the analog signal into a digital signal using an internal ADC (Analog-to-Digital Converter). This digital data is then ready to be sent to the microcontroller.

Step 5: Microcontroller Reads and Processes Data

The microcontroller (Arduino/ESP32) receives the digital data from the HX711 module through its input pins. It processes this data to calculate the actual weight using calibration values. The system continuously monitors this weight in real time.

Step 6: Display Weight on LCD

The calculated weight is displayed on the LCD screen. The LCD shows:

- Current weight of the products
- Status message (Stock OK / Stock LOW)

This allows users to easily monitor the system without needing a computer.

Step 7: Compare with Threshold Value

The system compares the measured weight with a predefined threshold value. This threshold represents the minimum acceptable stock level.

- If weight $\geq$ threshold $\rightarrow$ Stock is sufficient
- If weight $<$ threshold $\rightarrow$ Stock is low

Step 8: If Stock is LOW

When the weight falls below the threshold:

- **Buzzer Activation:**
The buzzer is turned ON to provide an audible alert indicating low stock.
- **Alert Notification (IoT):**
The WiFi module sends the data to a cloud platform (like ThingSpeak/Blynk).
Store managers can view this data remotely through a mobile app or web dashboard.
- **LCD Display:**
LCD shows “Stock LOW” message clearly.

Step 9: If Stock is SUFFICIENT

When the weight is above the threshold:

- Buzzer remains OFF
- LCD displays “Stock OK”
- System continues monitoring without interruption
- No alert is sent to cloud

10. CONCLUSION:

The Smart Retail Shelf system successfully demonstrates the use of IoT and embedded systems in modern retail management. It automates inventory monitoring, reduces human effort, and ensures real-time tracking of stock levels. The system minimizes losses due to stockouts and improves customer satisfaction.

In the future, this system can be enhanced by integrating:

- AI-based demand prediction
- RFID for product identification
- Mobile app notifications
- Data analytics for sales trends

Thus, the Smart Retail Shelf represents an important step towards smart retail automation and digital transformation.